

System Design: Notification System

Multi-channel Push/Email/SMS — Senior Engineer Interview Walkthrough

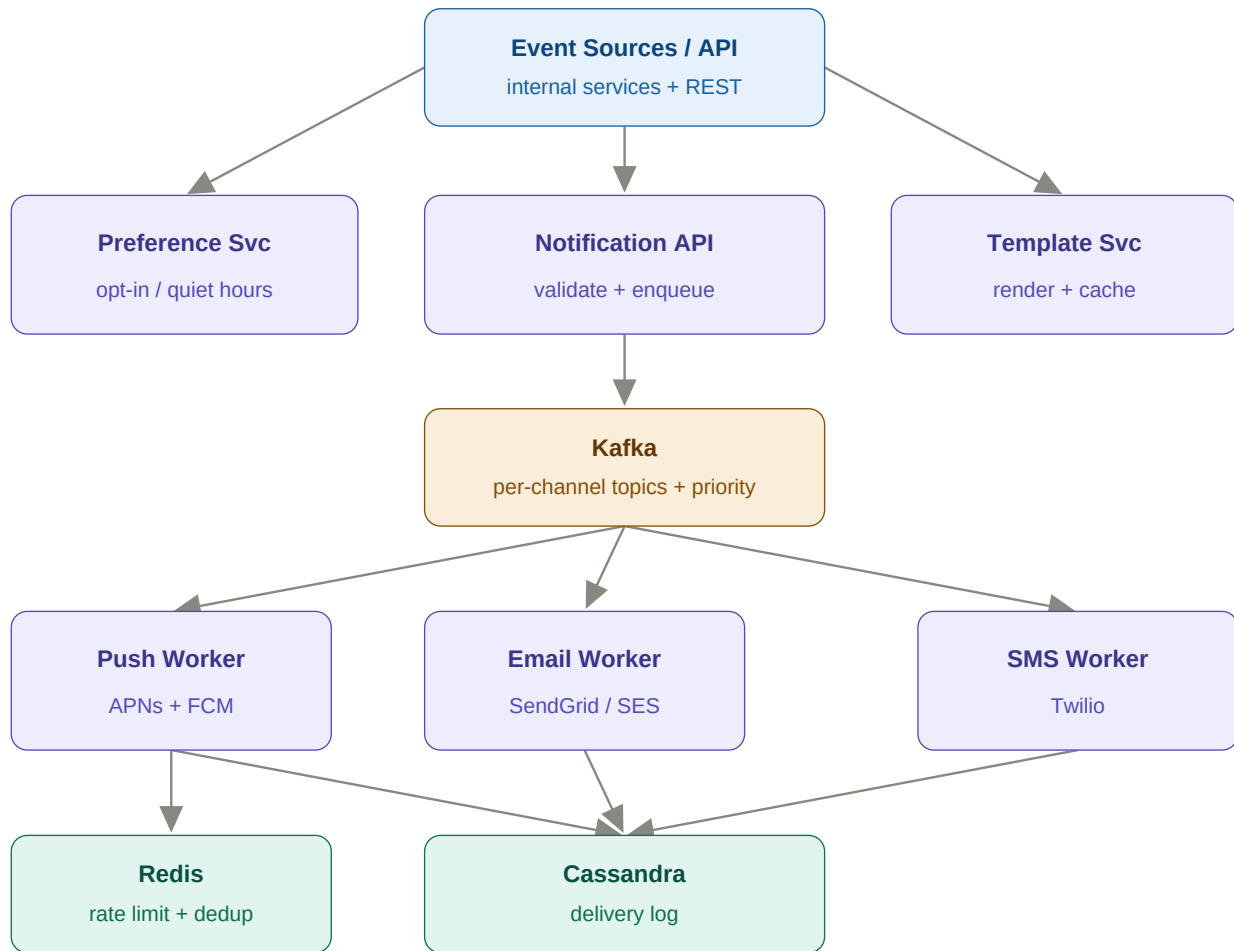
1. Requirements & Capacity

Think out loud: "Which channels: push, email, SMS, in-app? Who triggers: internal services or external API? Real-time or batch? I'll assume all 4 channels, both internal + API, real-time push, batched email digest."

Scale	Users/RPS	Architecture	Storage
Low	10k/day ~0.1 RPS	Monolith → direct APNs/FCM + Postgres log	<10 GB/yr
Mid	10M/day ~150 RPS	Notification service + Redis queue + provider adapters	~100 GB/yr
High	200M/day ~2.3k RPS peak 23k	Kafka per channel + priority queues + Cassandra log	~10 TB/yr

2. Architecture Diagram

Think out loud: "I decompose into: ingestion API, preference check, template engine, per-channel Kafka topics, channel workers, delivery tracker. Kafka decouples ingestion from dispatch — critical for absorbing traffic spikes."



Architecture: one Kafka topic per channel. Priority topics (HIGH=OTP, LOW=marketing) ensure password resets are never delayed by promotional blasts.

3. DB Schema

notification_log		
PK	notification_id	<i>UUID</i>
	user_id	<i>BIGINT</i>
	channel	<i>ENUM</i>
	template_id	<i>VARCHAR</i>
	status	<i>ENUM</i>
	created_at	<i>TIMESTAMP</i>
	sent_at	<i>TIMESTAMP</i>
	provider_msg_id	<i>VARCHAR</i>

user_preferences		
PK	user_id	<i>BIGINT</i>
PK	channel	<i>ENUM</i>
PK	notif_type	<i>VARCHAR</i>
	opted_in	<i>BOOLEAN</i>
	updated_at	<i>TIMESTAMP</i>

templates		
PK	template_id	<i>VARCHAR</i>
	channel	<i>ENUM</i>
	subject	<i>TEXT</i>
	body_template	<i>TEXT</i>
	locale	<i>VARCHAR(5)</i>

Schema. **Why this choice:** notification_log partition key = (user_id, channel). Query "show me last 50 notifications" = single partition read, O(1). At 200M events/day, Cassandra handles this where Postgres cannot.

4. API & Trade-offs

POST	/api/v1/notifications/send	{user_id, template_id, params, channel?, priority?}
POST	/api/v1/notifications/bulk	{user_ids[], template_id, params} — fan-out to segment
GET	/api/v1/notifications/{user_id}?limit=50	In-app notification inbox
PATCH	/api/v1/preferences/{user_id}	Update channel/type opt-in

Decision	Choice	Gain	Trade-off
Queue	Kafka per channel	Replay + partition isolation	Ops complexity
Log	Cassandra	Time-series petabyte scale	No complex queries

Prefs	Redis cache	<1ms pref check	Stale for 5 min
Delivery	At-least-once	Resilient	Dedup layer required